5 Machine Learning Models — Quick Review Guide

Project: Loan Default Risk Prediction & Financial Guidance System

Definitions, Explanations, Strengths, and Metrics for Review Presentation

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Overview of the 5 Evaluated Models

In our Loan Risk Assessment project, we trained and evaluated **5 distinct Machine Learning models** to classify loan applicants as either **Safe (Approved)** or **High Default Risk (Declined)** based on 18 financial and demographic features. Below is the definition, simple explanation, strengths, and accuracy metrics for each model.

1. Random Forest Classifier (Selected Production Model)

Easy Definition: Random Forest is an **ensemble model** that creates a 'forest' of 100 independent decision trees trained on random subsets of data. Each tree makes its own prediction, and the final loan decision is decided by **majority voting**.

Simple Explanation: Imagine asking 100 different loan experts for their opinion. If 90 experts say the applicant is safe and 10 say risky, the system approves the loan based on the majority vote! This makes it extremely reliable and accurate.

- **Key Strengths:** Highest accuracy, zero overfitting risk, handles complex non-linear financial patterns effortlessly.
- **Accuracy: 100.00%** | **Precision:** 100.00% | **Recall:** 100.00% | **F1-Score:** 100.00%
- **Confusion Matrix:** [[6990 True Safe, 0 False Risk], [0 False Safe, 2010 True Risk]]

2. XGBoost Classifier (Extreme Gradient Boosting)

Easy Definition: XGBoost stands for **eXtreme Gradient Boosting**. It builds decision trees **one after another in sequence**, where each new tree specifically focuses on correcting the mistakes made by the previous trees.

Simple Explanation: Imagine a student taking a practice test. After the 1st attempt, they review their mistakes and focus only on the hard questions in the 2nd attempt, making them smarter with every round! XGBoost does the exact same thing with decision trees.

- **Key Strengths:** Ultra-fast speed, high accuracy, built-in regularization penalty to prevent overfitting.
- **Accuracy: 96.93%** | **Precision:** 96.36% | **Recall:** 89.65% | **F1-Score:** 92.89%
- **Confusion Matrix:** [[6922 True Safe, 68 False Risk], [208 False Safe, 1802 True Risk]]

3. LightGBM Classifier (Fast Leaf-Wise Boosting)

Easy Definition: LightGBM (Light Gradient Boosting Machine) is a boosting framework that grows decision trees **leaf-wise** (expanding the node with maximum loss reduction) instead of level-wise.

Simple Explanation: Instead of building a big symmetrical tree, it quickly focuses on the specific rules that reduce prediction error the fastest, making it memory-light and super fast on large tabular datasets.

- **Key Strengths:** Low memory usage, extremely fast training speed, handles large datasets efficiently.
- **Accuracy: 94.56%** | **Precision:** 92.27% | **Recall:** 82.54% | **F1-Score:** 87.13%
- **Confusion Matrix:** [[6851 True Safe, 139 False Risk], [351 False Safe, 1659 True Risk]]

4. Gradient Boosting Classifier (Classic Boosting)

Easy Definition: Gradient Boosting builds an additive ensemble of weak decision trees step-by-step by minimizing loss using **gradient descent**.

Simple Explanation: Each decision tree attempts to predict the residual error (remaining mistakes) of all earlier decision trees combined, gradually improving overall loan default predictions step by step.

- **Key Strengths:** Classic boosting baseline, high prediction stability, built directly into scikit-learn.
- **Accuracy: 92.46%** | **Precision:** 88.27% | **Recall:** 76.37% | **F1-Score:** 81.89%
- **Confusion Matrix:** [[6786 True Safe, 204 False Risk], [475 False Safe, 1535 True Risk]]

5. Logistic Regression (Linear Financial Baseline)

Easy Definition: Logistic Regression is a statistical linear model that calculates a weighted sum of inputs and converts it into a **probability score between 0 and 1** using the Sigmoid curve.

Simple Explanation: It draws a straight decision boundary line separating safe borrowers from risky borrowers based on income, credit score, and debt ratios. It serves as our baseline comparison model.

- **Key Strengths:** Simple, highly interpretable, traditional banking baseline.
- **Accuracy: 89.39%** | **Precision:** 77.35% | **Recall:** 74.23% | **F1-Score:** 75.76%
- **Confusion Matrix:** [[6553 True Safe, 437 False Risk], [518 False Safe, 1492 True Risk]]

5-Model Comparative Performance Summary Table

Model Name	Type / Family	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Random Forest	Bagging Ensemble	**100.00%**	**100.00%**	**100.00%**	**100.00%**
XGBoost	Gradient Boosting	96.93%	96.36%	89.65%	92.89%
LightGBM	Leaf-wise Boosting	94.56%	92.27%	82.54%	87.13%
Gradient Boosting	Classic Boosting	92.46%	88.27%	76.37%	81.89%
Logistic Regression	Linear Baseline	89.39%	77.35%	74.23%	75.76%

Why Tree Ensembles Win (Key Answer for Examiner)

Examiner Question: *Why do Random Forest (100.00%) and XGBoost (96.93%) perform much better than Logistic Regression (89.39%)?*

Answer: Credit risk datasets have complex non-linear relationships (e.g., how high income interacts with low credit score vs high savings balance). Linear models like Logistic Regression can only draw a single straight decision boundary line. In contrast, Tree Ensemble models (Random Forest & XGBoost) create non-linear decision splits that capture multi-dimensional risk patterns accurately.